

Nathan Brewer

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EDUCATION

Swarthmore College | Major GPA: 3.96

Expected Graduation 2028

- Bachelor of Science in Engineering, Minor in Applied Mathematics
- Relevant Coursework: Multivariable Calculus, Differential Equations, Statics, Linear Physical Systems Analysis, Principles of Computer Engineering, Electrical Circuit Analysis
- Honors and Memberships: Sigma Xi, American Society of Mechanical Engineers

WORK EXPERIENCE

Research and Development Intern

May 2026 - August 2026

Ryzing Technologies, Staunton, VA

- Led design, physical construction/fabrication, iteration, and creation of technical drawing packages for programs including US Army Arctic Medical Operations and NASA Reentry Shielding
- Validated roll MOI of USMC ACV model to within 1% of datasheet specifications using MATLAB analysis and lead-weight ballasting, enabling wave pool testing
- Designed, fabricated, and installed control system to enhance in-house extreme wind testing setup, extending test capabilities, repeatability, and accuracy
- Developed supplier evaluation matrices and coordinated directly with vendor representatives to source EMI shielding fabric for a secure communications program

Engineering Research Associate

October 2024 - Present

Swarthmore College, Swarthmore, PA

- Created a biomimetic robotic barnacle nauplius model for exploration of subsurface ocean environments
- Designed and fabricated a light-activated analog motor control circuit with adjustable sensitivity, regulated output voltage, and integrated comparator/MOSFET switching
- Modeled and fabricated a compact clockwork arm mechanism using Fusion 360, optimizing pivot geometries to improve hydrodynamics and mechanical efficiency
- Achieved a 20% reduction in overall system size, integrating full system into a biologically accurate carapace for underwater testing at the University of Washington
- Currently working to improve system robustness and reliability in high-viscosity environments

Physics Teaching Assistant

September 2025 - December 2025

Swarthmore College, Swarthmore, PA

- Selected as a student teaching assistant for 60+ students in General Physics I (Mechanics)
- Attended all class sessions, coordinating in-class activities and providing one-on-one support
- Ran weekly 90-minute help sessions, assisting students with test prep and problem sets
- Attended biweekly pedagogy sessions to improve teaching skills and methods

EXTRACURRICULAR ACTIVITIES

Swarthmore Rocketry

- Founded and lead Swarthmore Rocketry, re-establishing the program and organizing the team
- Designed and fabricated 768-pixel thermal tracking module to control fins
- Model rocket designs in SolidWorks and OpenRocket to determine stability and flight characteristics
- Use MIG/TIG welding, manual machining, and carbon fiber fabrication to create custom components

Swarthmore Robotics

- Cofounded and lead Swarthmore Robotics, directly managing combat robotics sub-team
- Use CAD and FEA software to design components and run impact simulations
- Designed and fabricated modular antweight class BattleBots to introduce new members to design principles
- Leading development and fabrication of NHRL-compliant Beetleweight (3 lb.) combat robot

SKILLS

Software: ANSYS Mechanical, OpenRocket, MultiSim

CAD: SolidWorks, Fusion 360, Onshape, Siemens NX, KiCAD

Computer Languages: MATLAB, Python, SQL, CSS, HTML

Fabrication: MIG/TIG welding, manual machining, basic carpentry, 3D printing, soldering